

ARM-450 rev I.1

Kinematics: DH table, forward and inverse kinematics, Jacobian, workspace

in the Modified (Craig) DH convention with the velocity-propagation Jacobian -- after the myCobot 280 thesis and Rsine

Geometry taken from the rev I.1 CAD (servo axes of the assembled arm, tool-flange face):

$d_1 = 90.000$ $a_2 = 119.000$ $d_4 = 221.369$ $d_T = 88.089$ mm; J4, J5, J6 meet in one point (spherical wrist)

Forward kinematics = the CAD to $2.98e-13$ mm and the URDF (ikpy) to $2.55e-13$ mm over 2000 random poses

Jacobian by velocity propagation = the z x r form to $2.27e-13$; = central differences to $2.91e-10$ (relative)

Inverse kinematics in CLOSED FORM: up to 8 branches, exact to $2.70e-13$ mm on 5000 random poses

Reach: 405.5 mm from the J1 axis, 428.5 mm from the shoulder; tool face -49.5 .. 518.4 mm

Pen STRAIGHT DOWN is impossible: J2 + J3 + J5 pitch at most 172.5 deg; a table needs a tilted pen

Sine traced on a vertical board and on the table ring: closed form exact, DLS and ikpy re-trace to $1e-6$.. $1e-4$ mm

1 The convention: Modified (Craig) DH

Frame $\{i\}$ is attached to link i . Frames use only X and Z: Z_i along joint axis i (right-hand rule, +Z up at home -- here the vertical joints J1, J4, J6 point +z and the pitch joints J2, J3, J5 point +y, so a positive angle leans the arm forward / turns it counter-clockwise seen from above); X_i along the common normal from Z_i to Z_{i+1} ; Y_i completes the right-handed set.

a_{i-1} link length: the mutual-perpendicular distance between Z_{i-1} and Z_i , measured along X_{i-1}

α_{i-1} link twist: the angle from Z_{i-1} to Z_i , measured about X_{i-1}

d_i link offset: the distance from X_{i-1} to X_i , measured along Z_i (prismatic variable)

θ_i joint angle: the angle from X_{i-1} to X_i , measured about Z_i (revolute variable)

$${}^{i-1}_i T = Rot_x(\alpha_{i-1}) Trans_x(a_{i-1}) Rot_z(\theta_i) Trans_z(d_i)$$

$${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

arm450kin/robot.py: `craig_T()` is exactly this matrix; the analysis derives the table from the CAD with these rules (analysis/s1_dh_from_cad.py).

2 The DH table of ARM-450 rev I.1 (derived from the CAD)

ARM-450 rev I.1 -- Modified (Craig) DH parameters

i	α_{i-1} (deg)	a_{i-1} (mm)	d_i (mm)	θ_i	θ_i from servo q_i	servo range	joint
1	0	0.000	90.000	θ_1	$\theta_1 = -q_1$	$q_1 \pm 90^\circ$	J1 base yaw
2	-90	0.000	0.000	θ_2 (home -90°)	$\theta_2 = -q_2 - 90^\circ$	$q_2 \pm 54^\circ$	J2 shoulder
3	0	119.000	0.000	θ_3 (home $+90^\circ$)	$\theta_3 = q_3 + 90^\circ$	$q_3 \pm 72^\circ$	J3 elbow
4	90	0.000	221.369	θ_4	$\theta_4 = -q_4$	$q_4 \pm 90^\circ$	J4 forearm roll
5	-90	0.000	0.000	θ_5	$\theta_5 = q_5$	$q_5 \pm 46.5^\circ$	J5 wrist pitch
6	90	0.000	0.000	θ_6	$\theta_6 = -q_6$	$q_6 \pm 90^\circ$	J6 tool roll

$${}^{i-1}_i T = Rot_x(\alpha_{i-1}) Trans_x(a_{i-1}) Rot_z(\theta_i) Trans_z(d_i) \quad {}^6_7 T = Trans_z(d_T), \quad d_T = 88.089 \text{ mm}$$

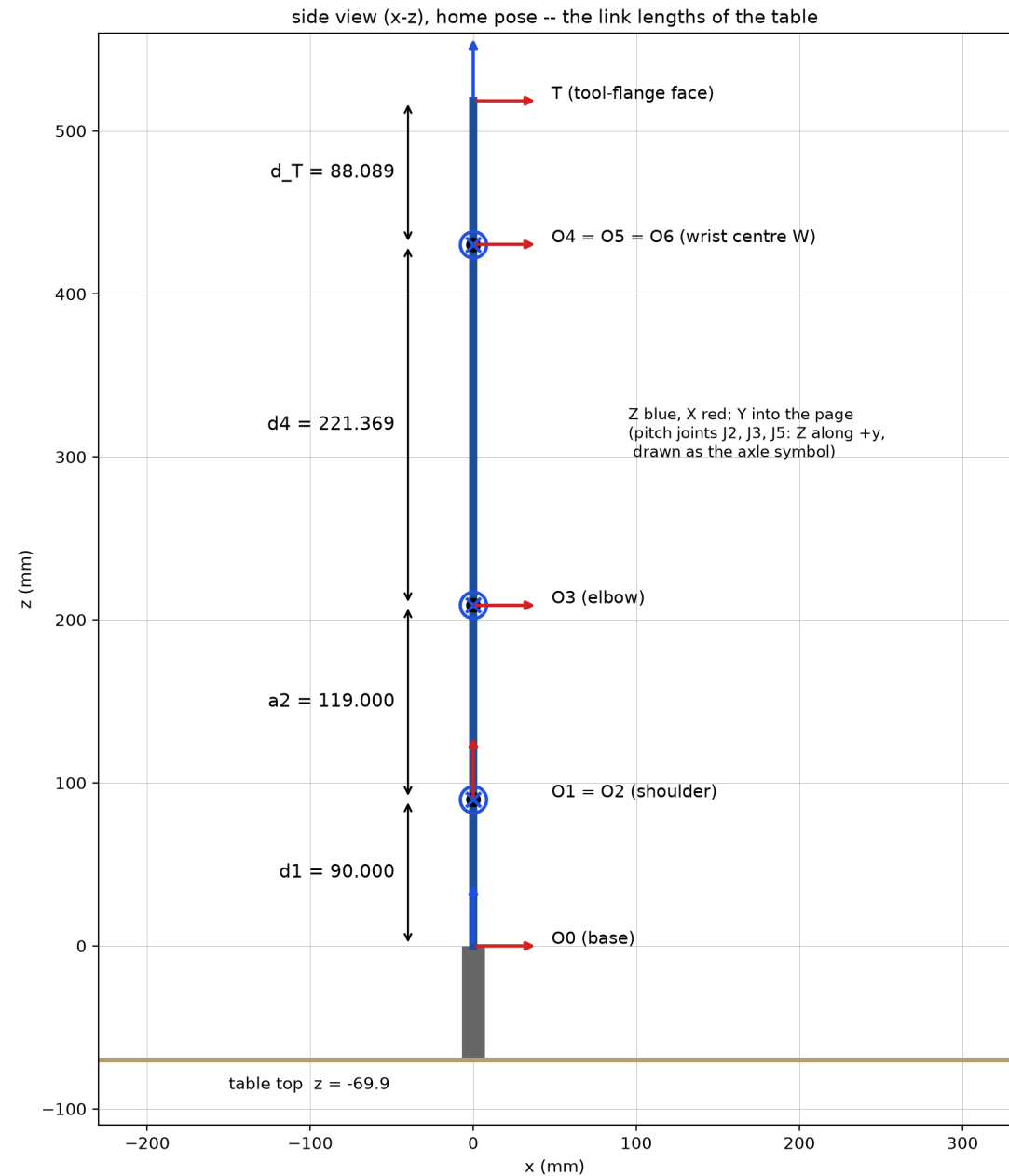
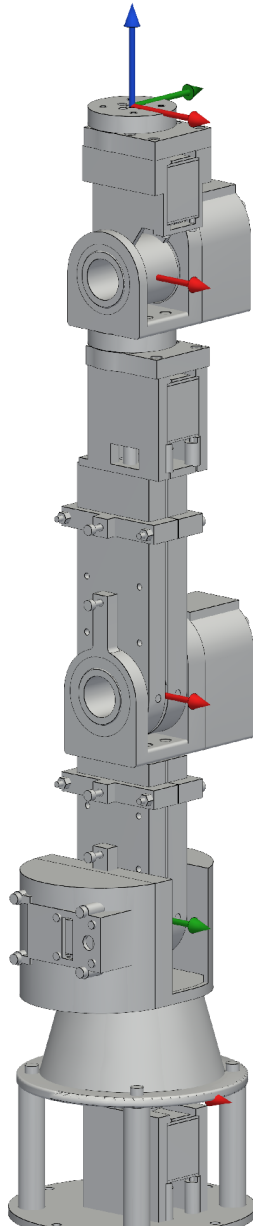
$${}^0_7 T = {}^0_1 T {}^1_2 T {}^2_3 T {}^3_4 T {}^4_5 T {}^5_6 T {}^6_7 T \quad \text{home: } {}^0_7 T = Trans_z(518.458)$$

Derived from the CAD servo axes and checked: every link transform has the exact Craig form; the frames match the model's to 1e-14 mm.
Spherical wrist (J4, J5, J6 meet at W, 430.369 mm above the base) -> closed-form IK.

2 Frames on the CAD and the link lengths

ARM-450 rev I.1 -- Modified (Craig) DH frames, derived from the CAD (home: all servos at 0)

frames on the CAD: X red, Y green, Z blue (O0 base, O1=O2 shoulder, O3 elbow, O4=O5=O6 wrist centre, T tool face)



3 Forward kinematics

$${}^0_7T(\theta) = {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6) {}^6_7T$$

$${}^6_7T = Trans_z(d_7), \quad d_7 = 88.089 \text{ mm (wrist centre to the tool-flange face)}$$

servo command q_i to DH angle: $\theta_i = s_i q_i + \theta_{i,0}$, $s = (-1, -1, +1, -1, +1, -1)$, $\theta_{i,0} = (0, -90^\circ, +90^\circ, 0, 0, 0)$

home (all $q_i = 0$, arm straight up): ${}^0_7T = Trans_z(518.458)$ -- tool frame parallel to the base

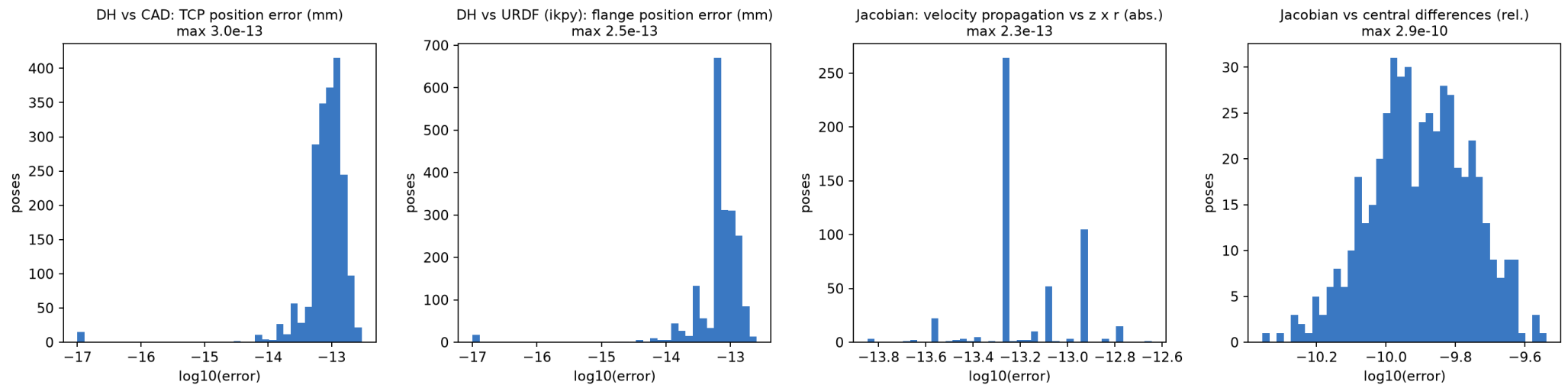
VERIFICATION (VERIFY_FK.log)

DH FK vs CAD (product of exponentials), position	max 2.98e-13 mm
DH FK vs CAD, orientation	max 1.49e-15 rad
DH FK vs URDF (ikpy 3.4.2), position	max 2.55e-13 mm
DH FK vs URDF, orientation	max 8.69e-16 rad
velocity-propagation J vs z_i x (p_e - p_i)	max 2.27e-13 mm/rad
velocity-propagation J vs central differences (relative)	max 2.91e-10 -

Three independent references: the CAD itself (the assembly moved by the product of exponentials of its own measured servo axes -- no DH anywhere), the URDF that RViz uses (through ikpy), and for the Jacobian the textbook z x r form and finite differences.

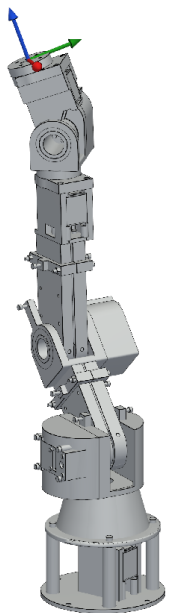
3 FK verified: the DH frame sits on the CAD's tool face in every pose

ARM-450 rev I.1 -- forward kinematics verified: DH vs the CAD, vs the URDF, Jacobian two ways

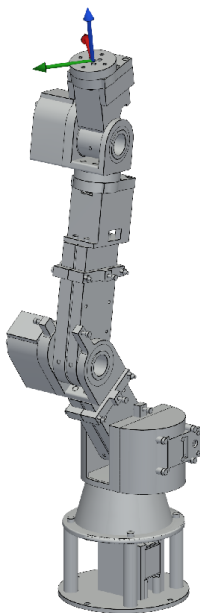


the CAD posed by its own servo axes; the red/green/blue frame is the DH prediction ${}^0_T T(q)$ -- it sits on the tool-flange face in every pose

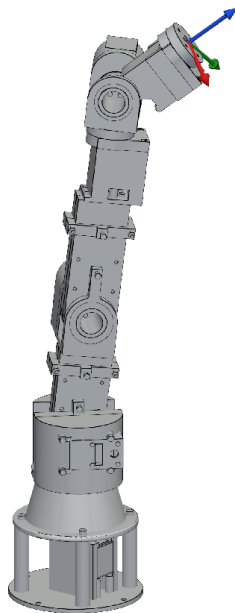
$q = (23, 43, 40, -49, -19, 67) \text{ deg}$



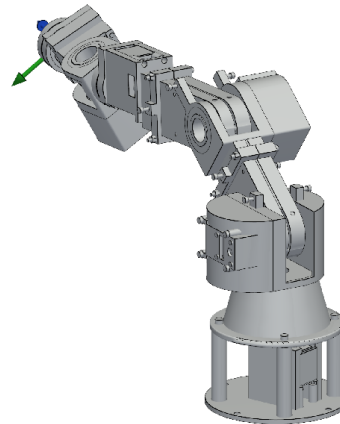
$q = (-89, 35, 43, -6, -18, -40) \text{ deg}$



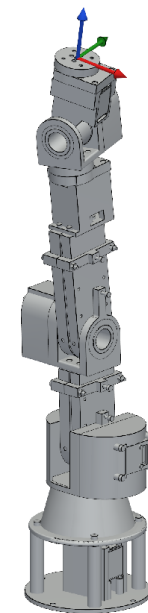
$q = (-44, -6, 1, 10, 46, 53) \text{ deg}$



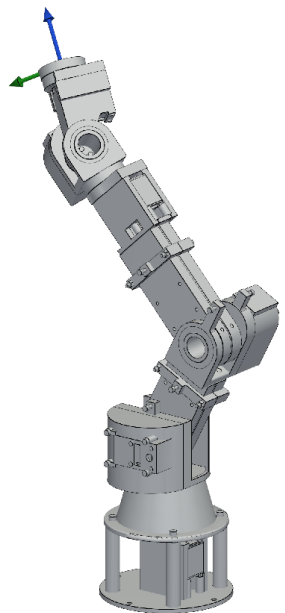
$q = (22, 53, -41, -61, 10, -82) \text{ deg}$



$q = (-84, 2, -5, 75, 12, 3) \text{ deg}$



$q = (-1, -27, -70, -55, 18, -54) \text{ deg}$



4 Jacobian by velocity propagation (Craig)

$${}^{i+1}\omega_{i+1} = {}^{i+1}R {}^i\omega_i + \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1}$$

$${}^{i+1}v_{i+1} = {}^{i+1}R ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1}) + \dot{d}_{i+1} {}^{i+1}\hat{Z}_{i+1}$$

revolute joint: drop the $\dot{d}$ term; prismatic: drop the $\dot{\theta}$ term (kept in the code)

$$\text{tool face: } {}^6v_T = {}^6v_6 + {}^6\omega_6 \times (0, 0, d_T)^T$$

$$\text{base frame: } {}^0v_T = {}^0R {}^6v_T, \quad {}^0\omega_6 = {}^0R {}^6\omega_6; \quad \text{column } k \text{ of } J = \text{the tip twist for } \dot{\theta}_k = 1$$

Implemented in arm450kin/robot.py jacobian(): each column is carried frame by frame from the base to the tip exactly as above, then expressed in the base frame. Checked against the geometric form $z_i \times (p_T - p_i)$; z_i and against central differences of the FK:

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FORWARD KINEMATICS + JACOBIAN VERIFICATION (2000 random poses inside the joint limits; Jacobian 500)
velocity-propagation J vs z_i x (p_e - p_i)                max 2.27e-13 mm/rad
velocity-propagation J vs central differences (relative)   max 2.91e-10 -
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manipulability (Yoshikawa) $w = \sqrt{\det(J_v J_v^T)}$, condition number $\kappa = \sigma_{max}/\sigma_{min}$

5 Inverse kinematics in closed form (spherical wrist, Pieper)

1. wrist centre: $W = p - d_7 {}^0R_7 \hat{z}$ (J4, J5, J6 meet at W)

2. base: $\theta_1 = \text{atan2}(W_y, W_x)$ or $+\pi$ (arm reaching back)

3. arm plane: $r = \pm \sqrt{W_x^2 + W_y^2}$, $s = W_z - d_1$, $D = \frac{r^2 + s^2 - a_2^2 - d_4^2}{2a_2d_4}$

$$\varphi_3 = \pm \arccos D, \quad \varphi_2 = \text{atan2}(r, s) - \text{atan2}(d_4 \sin \varphi_3, a_2 + d_4 \cos \varphi_3)$$

$$\theta_2 = \varphi_2 - 90^\circ, \quad \theta_3 = \varphi_3 + 90^\circ \quad (\varphi = \text{lean from vertical})$$

4. wrist: $\text{Rot}_x(90^\circ)^T {}^0R_3^T {}^0R_6 = \text{Rot}_z(\theta_4) \text{Rot}_y(\theta_5) \text{Rot}_z(\theta_6) = M$ (Z-Y-Z Euler)

$$\theta_5 = \text{atan2}(\pm \sqrt{M_{13}^2 + M_{23}^2}, M_{33}), \quad \theta_4 = \text{atan2}(\pm M_{23}, \pm M_{13}), \quad \theta_6 = \text{atan2}(\pm M_{32}, \mp M_{31})$$

5. 2 (shoulder) x 2 (elbow) x 2 (wrist) = up to 8 solutions; the joint limits keep the valid ones

(a) closed form, 5000 random poses inside the joint limits

worst FK(branch) - target, all 8 branches: 2.70e-13 mm

poses whose original configuration was NOT among the valid branches: 0

valid branches per pose (joint limits): mean 1.37, min 1, max 2; histogram [0, 3161, 1839]

(b) 200 of the poses, full pose (position + orientation), joint limits enforced

LOCAL -- seed = the true solution moved 3 deg per joint (what a traced path gives):

damped least squares (Rsine method): 190 / 200 converged to 1e-6 mm, 190 of them on a closed-form branch

ikpy on the URDF (cross-check): 200 / 200 within 0.01 mm + 1e-3 rad, median error 4.2e-07 mm

GLOBAL -- seed = the fixed ready pose $q = (0, 30, -60, 0, 30, 0)$ deg, target anywhere:

damped least squares: 78 / 200 (78 on a closed-form branch)

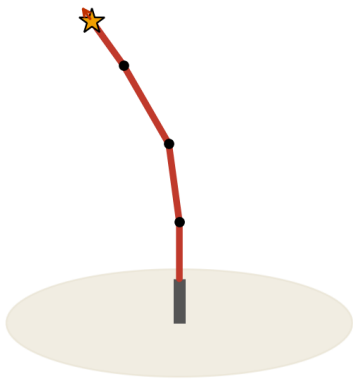
ikpy: 125 / 200, median error 3.48e-06 mm

-> numerical solvers are LOCAL: exact near a seed, unreliable across the workspace; the closed form has no seed and returns every branch -- use it, and the numerical ones along paths.

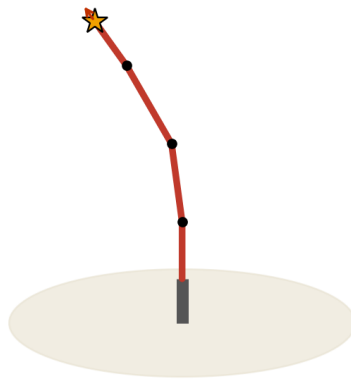
5 The 8 branches of one pose

Closed-form IK: the 8 branches for one tool pose (TCP [-285.8 165.6 307.5] mm) -- green = inside every joint limit

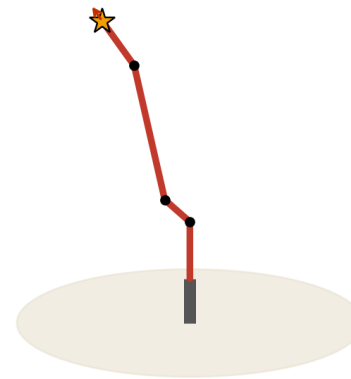
front elbow+ wrist+
 $q = (-145, -25, 50, -140, 30, -20)$ deg
outside the limits of J1,4



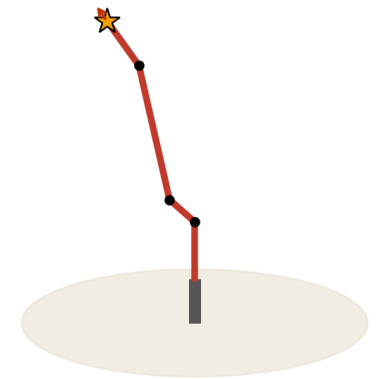
front elbow+ wrist-
 $q = (-145, -25, 50, 40, -30, 160)$ deg
outside the limits of J1,6



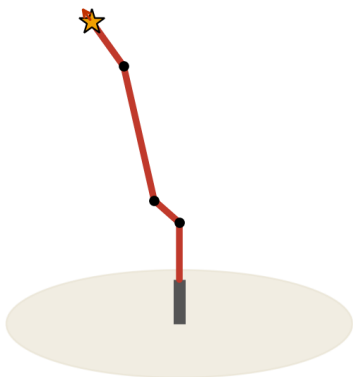
front elbow- wrist+
 $q = (-145, -91, -50, -63, 21, -103)$ deg
outside the limits of J1,2,6



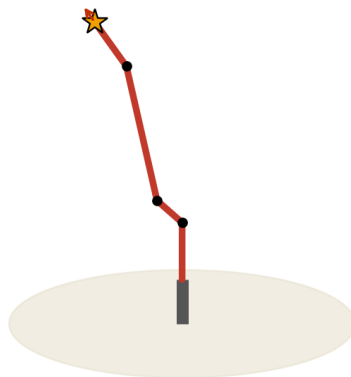
front elbow- wrist-
 $q = (-145, -91, -50, 117, -21, 77)$ deg
outside the limits of J1,2,4



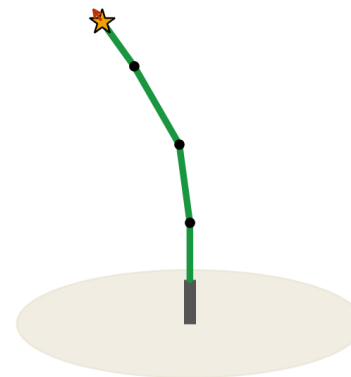
back elbow+ wrist+
 $q = (35, 91, 50, 117, 21, -103)$ deg
outside the limits of J2,4,6



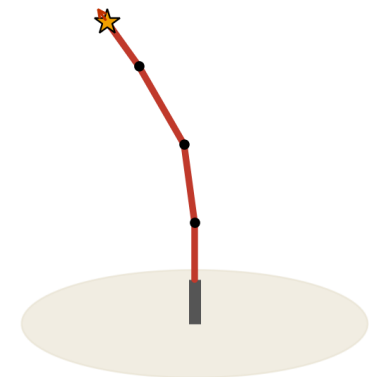
back elbow+ wrist-
 $q = (35, 91, 50, -63, -21, 77)$ deg
outside the limits of J2



back elbow- wrist+
 $q = (35, 25, -50, 40, 30, -20)$ deg
VALID



back elbow- wrist-
 $q = (35, 25, -50, -140, -30, 160)$ deg
outside the limits of J4,6



6 Numerical IK and the solver comparison (after the thesis, Exp. 2)

damped least squares (Rsine): $\Delta\theta = J^T(JJ^T + \lambda^2 I)^{-1}e$

pseudoinverse: $\Delta\theta = J^\dagger e$

Jacobian transpose: $\Delta\theta = \alpha J^T e$, $\alpha = \langle e, JJ^T e \rangle / \|JJ^T e\|^2$

Levenberg-Marquardt: $(J^T J + \lambda \text{diag}(J^T J)) \Delta\theta = J^T e$, λ adapted

error $e = (p^* - p, w \frac{1}{2} \sum_k x_k \times x_k^*)$, $w = 100$ mm/rad; joint limits clamped

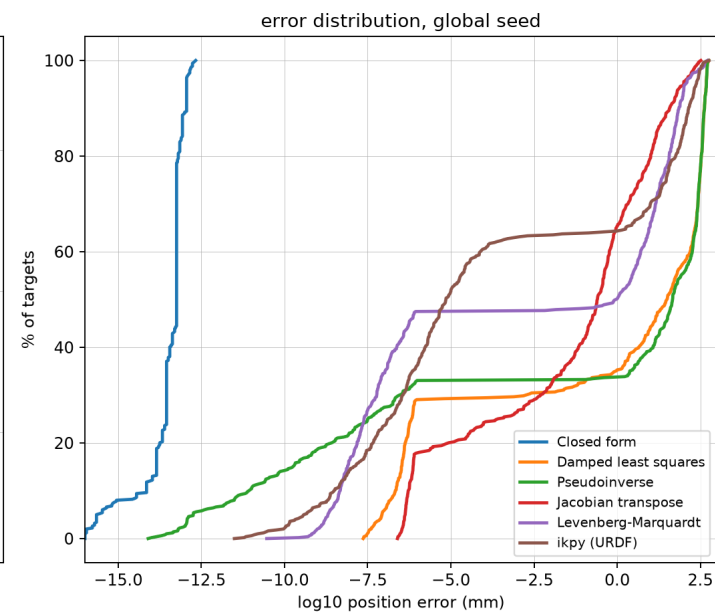
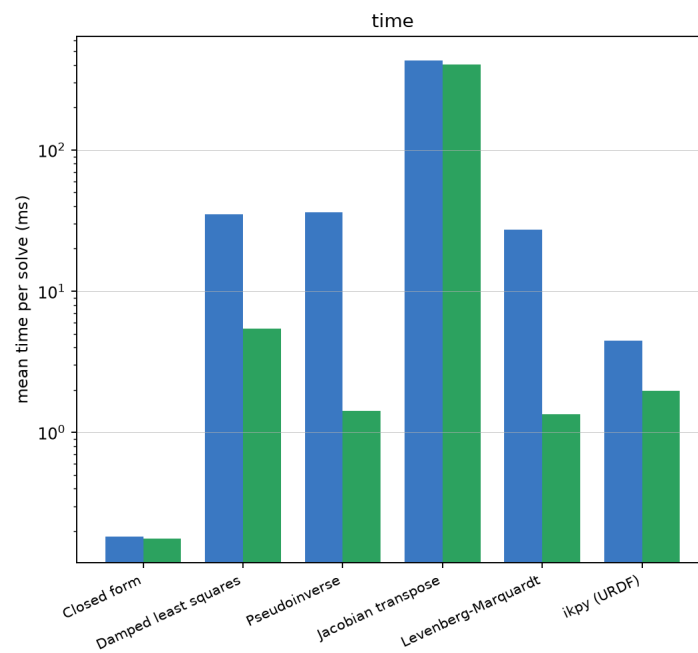
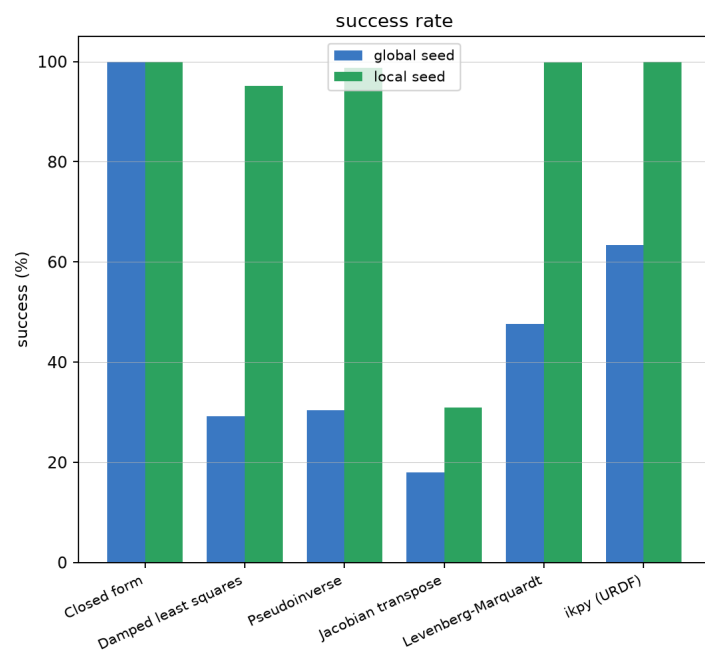
IK SOLVER COMPARISON -- 500 random reachable full poses, joint limits enforced

solver	seed	success	mean iter	mean ms	median err mm
Closed form	GLOBAL	100.0%	1.0	0.18	5.68e-14
Closed form	LOCAL	100.0%	1.0	0.18	5.68e-14
Damped least squares	GLOBAL	29.2%	221.6	34.89	3.12e+01
Damped least squares	LOCAL	95.2%	34.6	5.43	2.59e-07
Pseudoinverse	GLOBAL	30.4%	212.9	36.40	4.62e+01
Pseudoinverse	LOCAL	98.8%	8.7	1.42	1.09e-09
Jacobian transpose	GLOBAL	18.0%	2831.4	430.78	2.52e-01
Jacobian transpose	LOCAL	31.0%	2650.4	402.98	4.81e-04
Levenberg-Marquardt	GLOBAL	47.6%	165.5	27.15	9.10e-01
Levenberg-Marquardt	LOCAL	99.8%	8.6	1.35	5.89e-08
ikpy (URDF)	GLOBAL	63.4%	-	4.46	7.08e-06
ikpy (URDF)	LOCAL	100.0%	-	1.97	4.52e-07

GLOBAL = seed at the ready pose, LOCAL = seed 3 deg from the answer (a traced path).
The closed form needs no seed and is exact; the seed only picks among its valid branches.

6 Success, time and accuracy of the solvers

IK solvers on ARM-450 rev I.1 (after the myCobot thesis, Exp. 2): 500 random reachable full poses



7 Reachable workspace

self-checks: batch FK vs model FK 0.0e+00 mm; pen feasibility batch vs full IK (roll swept 5 deg) agree on 100.0 % of 400 pens (200 held by real poses + 200 random; 201 vs 201 feasible)

REACHABLE (tool-flange face, 400000 random poses inside the joint limits)

horizontal reach from the J1 axis: max 405.5 mm

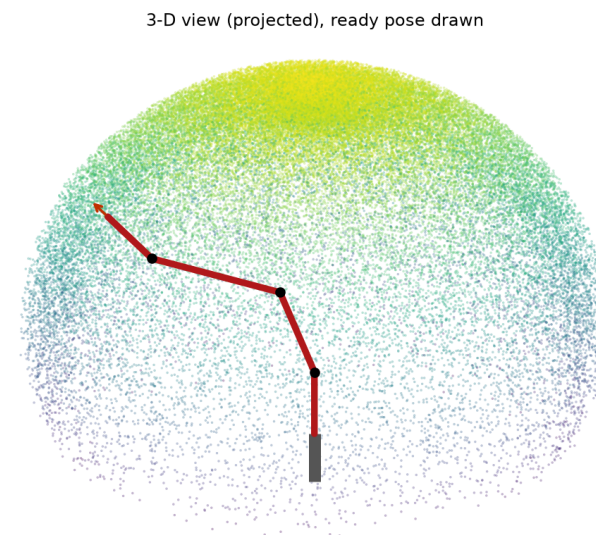
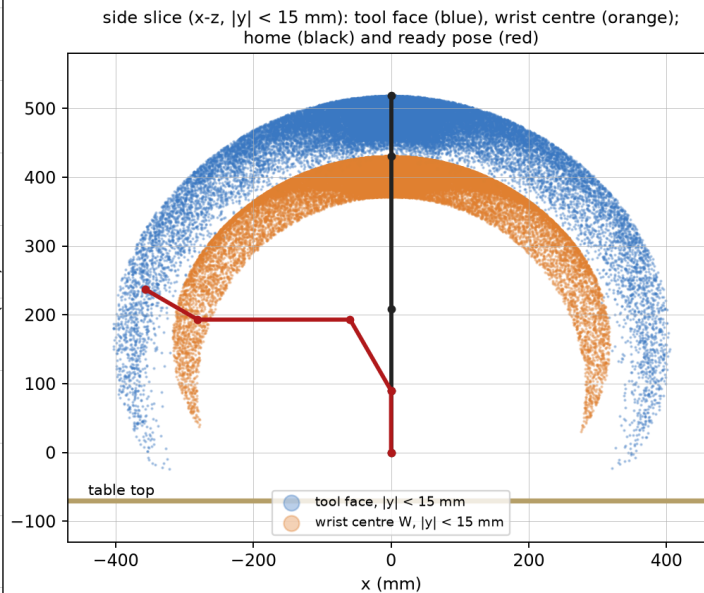
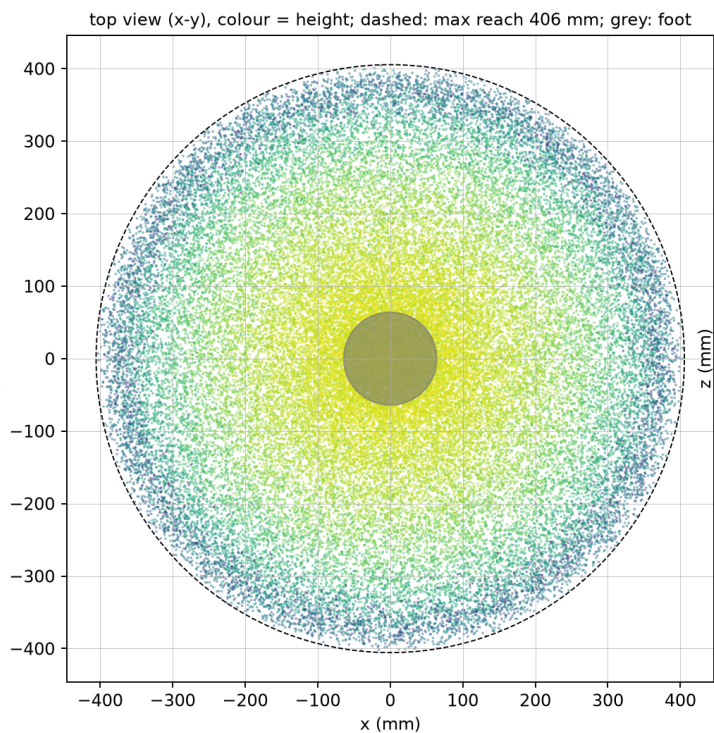
distance from the shoulder (0, 0, 90): max 428.5 mm ($a_2 + d_4 + d_T = 428.5$)

height: -49.5 .. 518.4 mm; below the table top (z < -69.9): 0.0 % of poses

volume above the table (occupied 10 mm voxels): 74.59 litres

wrist centre: r max 317.6 mm, z 30.5 .. 430.4 mm

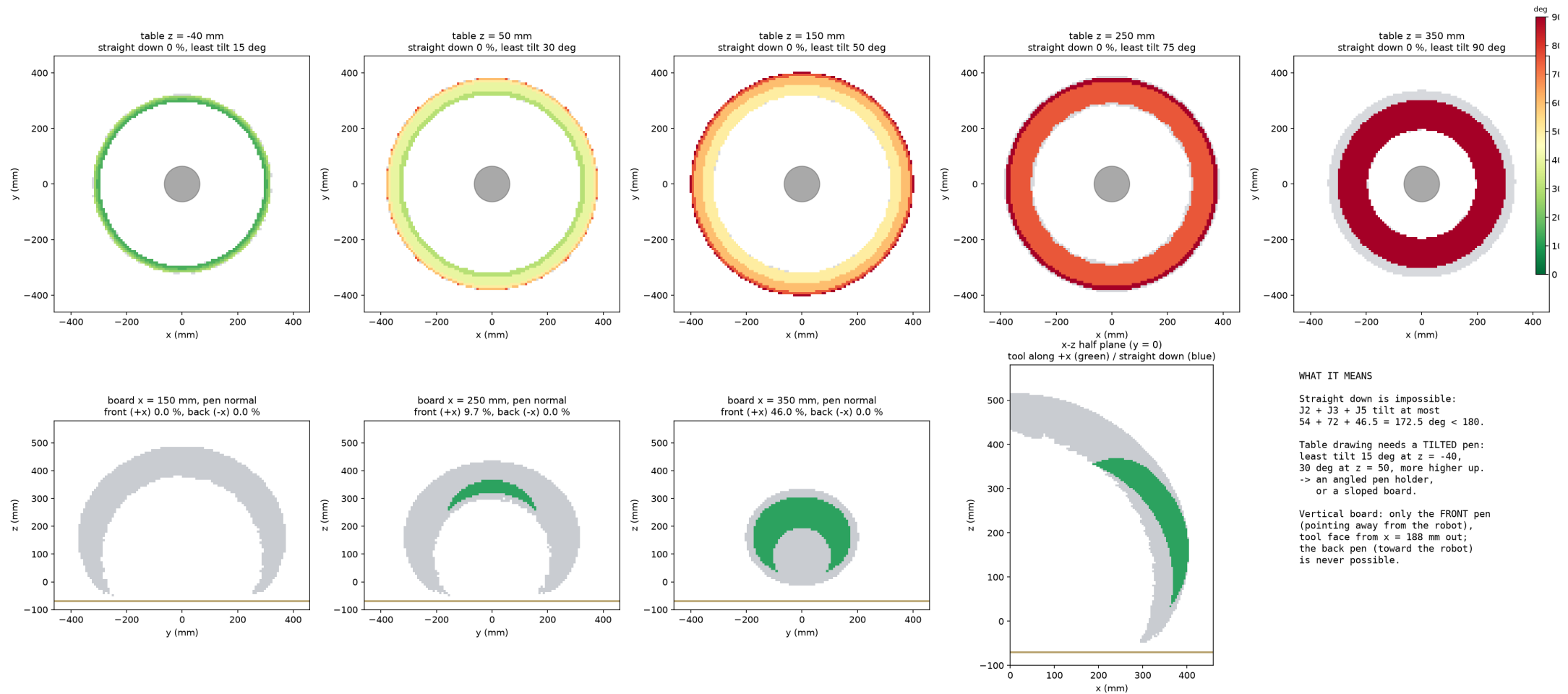
ARM-450 rev I.1 -- reachable workspace (tool-flange face, 400000 random poses, every joint inside its limits)



7 Task-constrained (drawable) workspace -- where the pen can stand normal to a surface

Task-constrained workspace (after the myCobot thesis): where the tool can stand NORMAL to a surface, roll free

top: horizontal tables -- colour = least pen tilt from straight down that some pose can hold (grey: reachable only); bottom: vertical boards -- green front pen (+x), blue back pen (-x). Exact per 8 mm cell.



7 What the task workspace says

TASK-CONSTRAINED (drawable) -- share of the reachable cells where the tool can stand normal to the surface

NOTE: straight down needs 180 deg of pitch; J2 + J3 + J5 give at most $54 + 72 + 46.5 = 172.5$ deg

horizontal z = -40.0 pen straight down: 0.0 % of 845 reachable cells; least tilt that works 15 deg; ≤ 30 deg on 94.2 %, ≤ 45 deg on 94.2 %

horizontal z = 50.0 pen straight down: 0.0 % of 2211 reachable cells; least tilt that works 30 deg; ≤ 30 deg on 24.6 %, ≤ 45 deg on 74.0 %

horizontal z = 150.0 pen straight down: 0.0 % of 3033 reachable cells; least tilt that works 50 deg; ≤ 30 deg on 0.0 %, ≤ 45 deg on 0.0 %

horizontal z = 250.0 pen straight down: 0.0 % of 3562 reachable cells; least tilt that works 75 deg; ≤ 30 deg on 0.0 %, ≤ 45 deg on 0.0 %

horizontal z = 350.0 pen straight down: 0.0 % of 3787 reachable cells; least tilt that works 90 deg; ≤ 30 deg on 0.0 %, ≤ 45 deg on 0.0 %

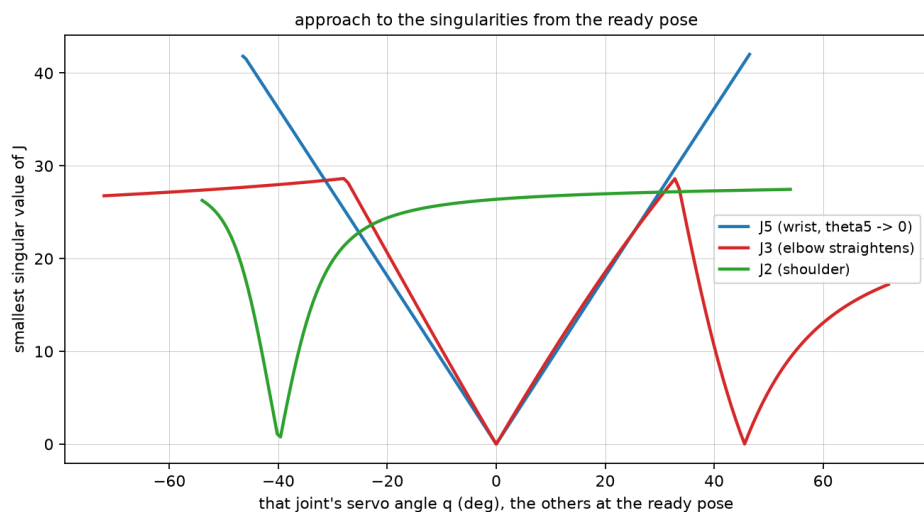
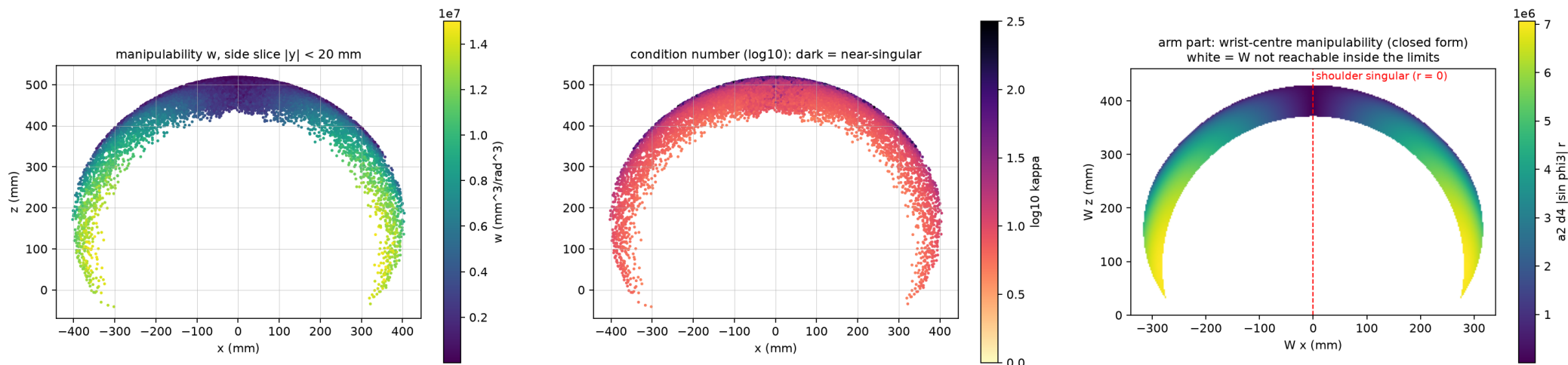
vertical x = 150.0 pen +x (front): 0.0 %, pen -x (back): 0.0 % of 2030 reachable cells

vertical x = 250.0 pen +x (front): 9.7 %, pen -x (back): 0.0 % of 2093 reachable cells

vertical x = 350.0 pen +x (front): 46.0 %, pen -x (back): 0.0 % of 1780 reachable cells

8 Manipulability and singularities (after the thesis, Exp. 3)

ARM-450 rev I.1 -- manipulability and conditioning (velocity-propagation Jacobian, 150000 random poses)



MANIPULABILITY + SINGULARITIES

arm manipulability $|\det J_W| = a_2 d_4 |\sin \phi_{i3}| r$: closed form vs velocity propagation, max rel. diff 2.0×10^{-13}

over 150000 random poses inside the joint limits (tool face, J_v):

manipulability w : median 4.68×10^6 , max $1.52 \times 10^7 \text{ mm}^3/\text{rad}^3$
 condition number κ : median 10.2, 95th percentile 34.5
 6-D min singular value (100 mm scaling) < 5 % of its median: 3.6 % of poses

SINGULARITIES (arm straight up at home is all three at once):

wrist $\theta_5 = 0$ J4 and J6 in line -- rank loss 1 (home, and anywhere $q_5 = 0$)
 elbow $\phi_{i3} = 0$ or 180 upper arm and forearm in line -- W on the reach sphere (home)
 shoulder $r = 0$ W on the J1 axis -- J1 moves nothing (home)
 rank of J : home 3, ready pose 6; smallest singular value home 3.03×10^{-31} , ready 27.2 (100 mm scaling)
 -> start and work from a bent pose (the manual's ready pose $q = 0, 30, -60, 0, 30, 0$ deg).

9 Sine tracing (after Rsine)

SINE TRACING (closed-form IK, continuity = nearest valid branch)

VERTICAL: board $x = 350$, pen normal $+x$ (away from the robot)

sine: centre (350, 0, 260) mm, width 200 mm, amplitude 33 mm, 2 cycles, 150 points

closed form: FK error max $1.3\text{e-}13$ mm, pen axis error $2.6\text{e-}16$; all points inside the joint limits: True

motion: largest joint step between points 2.82 deg; smallest margin to a limit 2.5 deg (pen roll free)

DLS re-trace (seed = previous point): max error $7.9\text{e-}07$ mm, mean 8.0 iterations

ikpy re-trace (URDF): max error $9.3\text{e-}05$ mm

TABLE: table $z = 25$, pen tilted 30 deg from straight down, leaning out along the radius

sine along an arc of radius 340 mm (the drawable ring), arc length 300 mm, radial amplitude 12 mm, 2 cycles, 150 points

closed form: FK error max $1.3\text{e-}13$ mm, pen axis error $2.9\text{e-}16$; all points inside the joint limits: True

motion: largest joint step between points 0.74 deg; smallest margin to a limit 1.0 deg (pen roll free)

DLS re-trace (seed = previous point): max error $6.2\text{e-}07$ mm, mean 4.4 iterations

ikpy re-trace (URDF): max error $1.9\text{e-}06$ mm

9 Sine tracing: paths, servo angles, errors

Sine tracing on ARM-450 rev I.1 (after Rsine): closed-form IK, placed with the task-constrained workspace

vertical board, pen +- 150 points, 4 of the poses

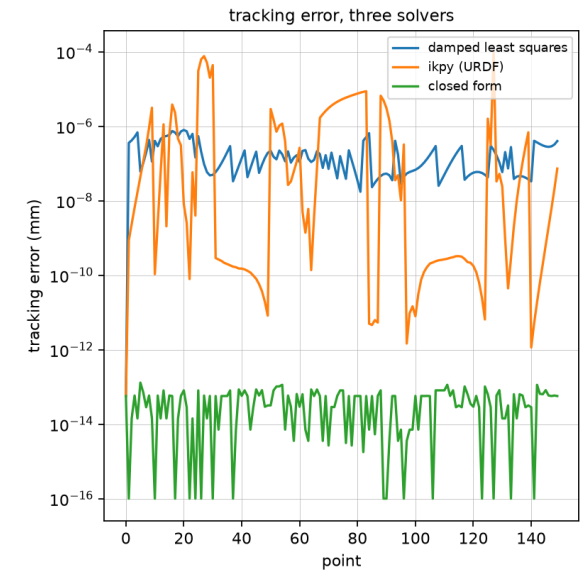
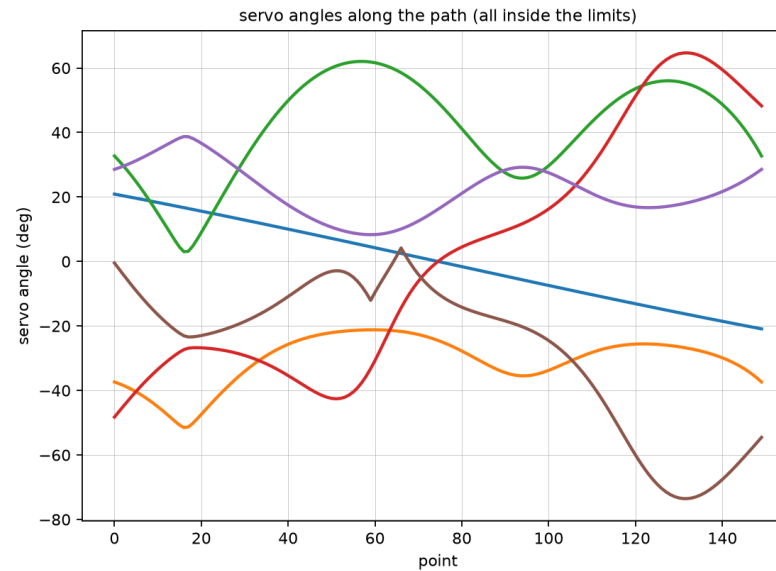
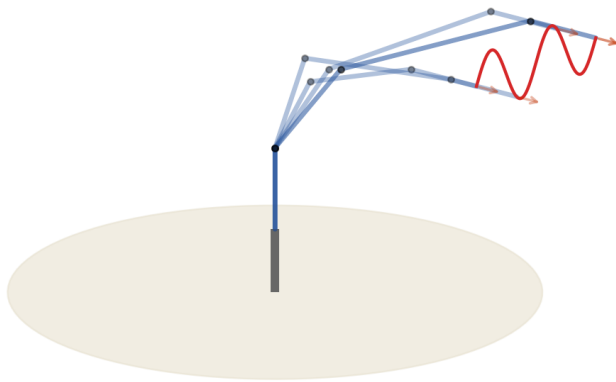
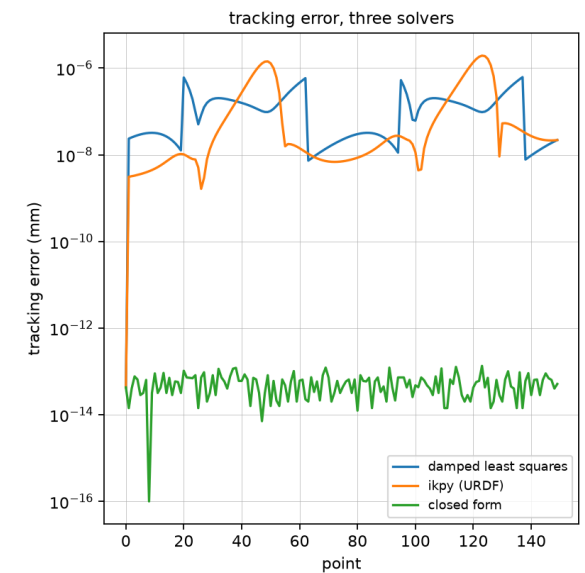
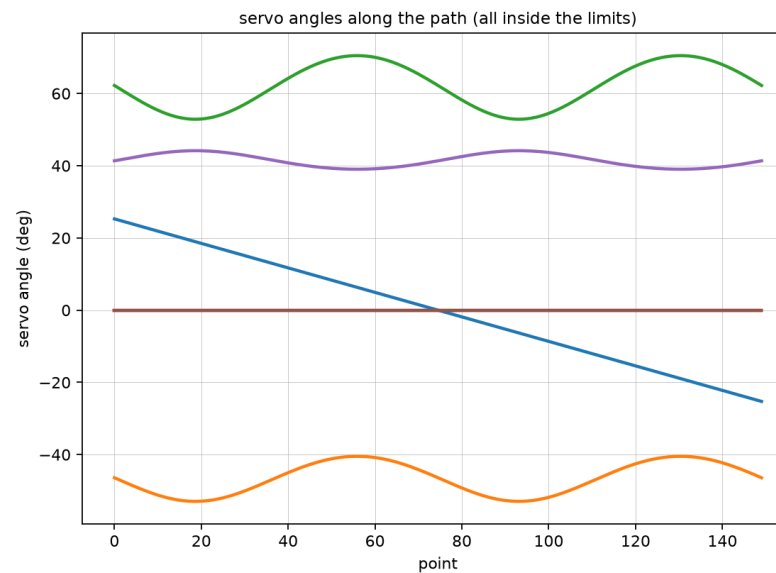
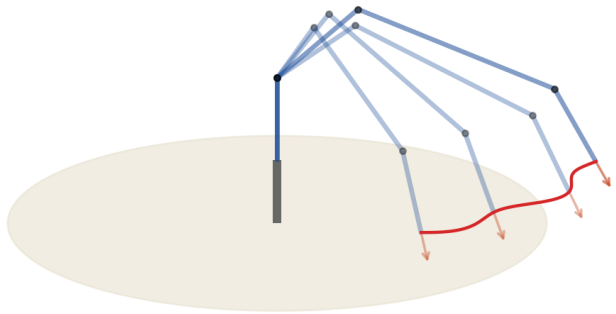
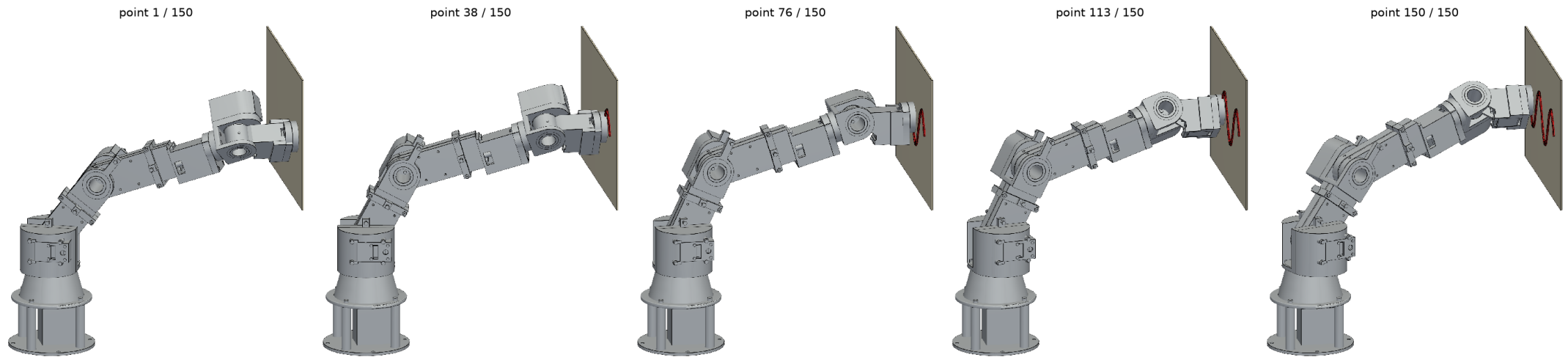


table z = 25, pen tilted 30 deg -- 150 points, 4 of the poses



9 The CAD arm drawing on the board (animations: figures/sine_vertical.gif, sine_tab

The CAD arm posed by the closed-form IK along the vertical-board sine: pen (black) normal to the board, the drawn curve in red



10 The sine as a timed trajectory on the ST3215 servos

a pen is symmetric: task = tool-face position + pen axis = 5-D, arm = 6 joints -> 1 redundant DOF

$$J_5 = [J_v; u^T J_\omega; w^T J_\omega] \quad (u, w \perp \text{pen axis}), \quad \Delta\theta = J_5^\dagger \Delta x_5 + (I - J_5^\dagger J_5) z$$

$z = -k(\theta - \theta_{mid})/\theta_{half}^2$ keeps the joints near the middle of their ranges; Newton correction per point

servo speed $\dot{q}_i = \frac{dq_i}{ds} v$ against the ST3215 rating $0.222 \text{ s}/60^\circ = 270^\circ/\text{s}$

encoder floor $|\delta x| \leq \sum_i \|J_{v,i}\| \frac{\Delta q}{2}, \quad \Delta q = 360^\circ/4096$

TIMED TRAJECTORIES + ENCODER RESOLUTION

ST3215 (Waveshare wiki): $0.222 \text{ s}/60 \text{ deg}$ at 12 V = 270 deg/s no-load; 4096 counts/rev = 0.0879 deg/count

VERTICAL sine (board x = 350): path 345 mm

resolved rate (5-D task, roll free, null space to the joint centres): error $9.3\text{e-}11 \text{ mm}$, pen axis $9.2\text{e-}13$; inside the limits: True

largest joint rate per mm of pen travel 1.65 deg/mm (joint 3); closed-form path of step 7: largest step 2.82 deg

pen speed at which a servo reaches its no-load rating: 164 mm/s ; at half the rating: 82 mm/s

one pass of the sine at half the rating: 4.2 s

encoder floor (every joint off by half a count): tool face $0.87 \dots 0.93 \text{ mm}$ along the path

TABLE sine (table z = 25, pen tilt 30 deg): path 318 mm

resolved rate (5-D task, roll free, null space to the joint centres): error $1.8\text{e-}13 \text{ mm}$, pen axis $5.0\text{e-}16$; inside the limits: True

largest joint rate per mm of pen travel 0.33 deg/mm (joint 3); closed-form path of step 7: largest step 0.74 deg

pen speed at which a servo reaches its no-load rating: 822 mm/s ; at half the rating: 411 mm/s

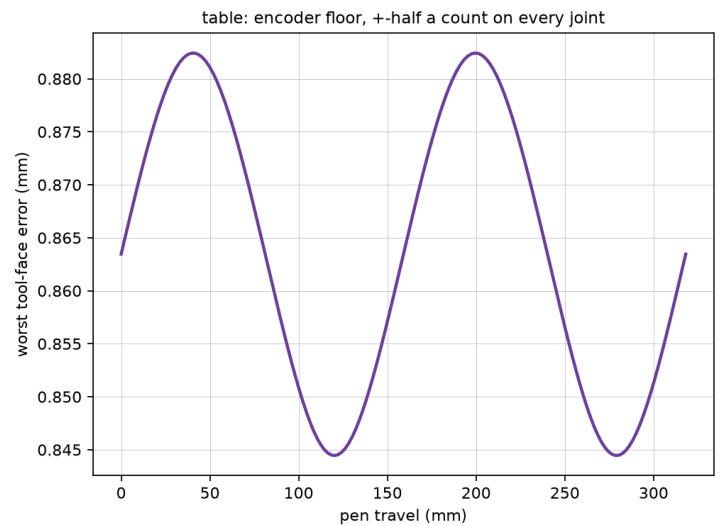
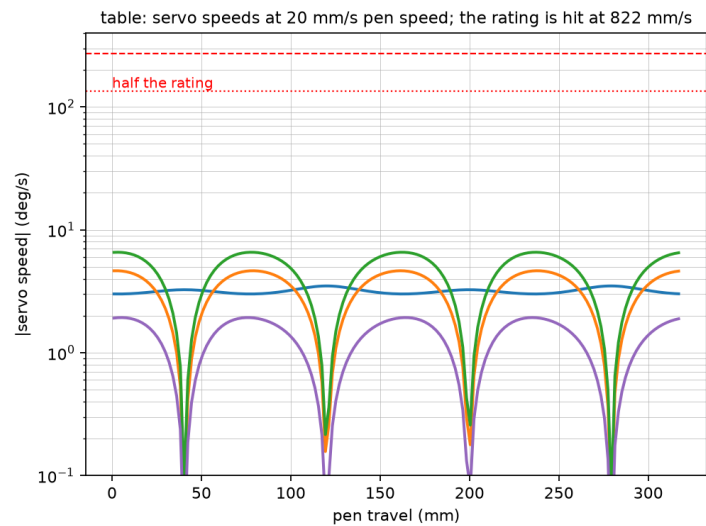
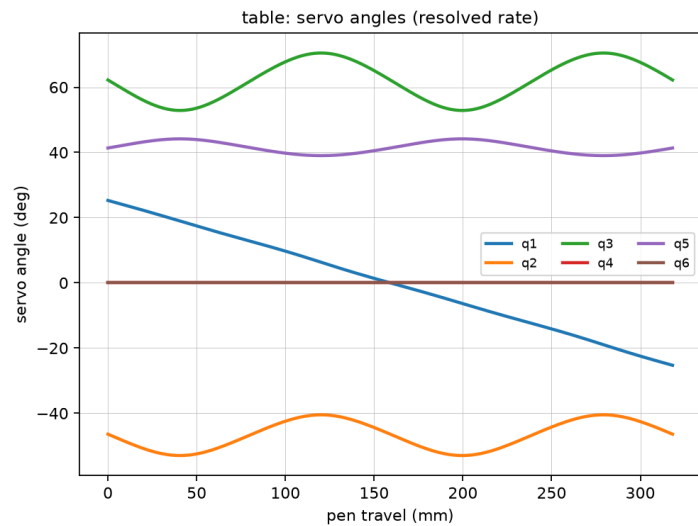
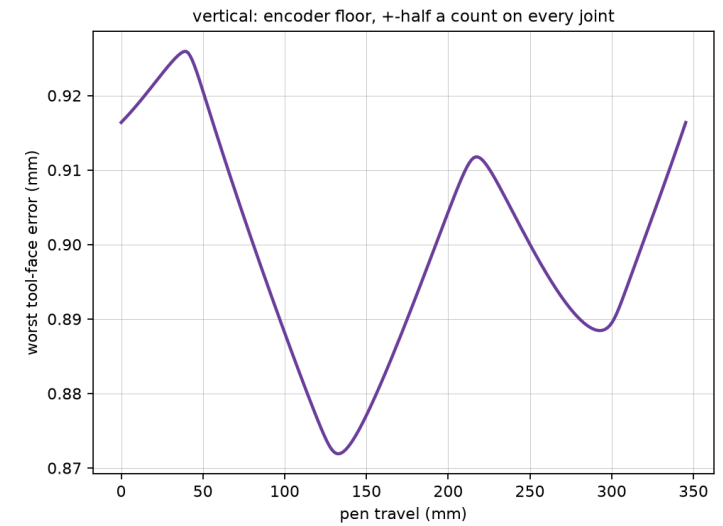
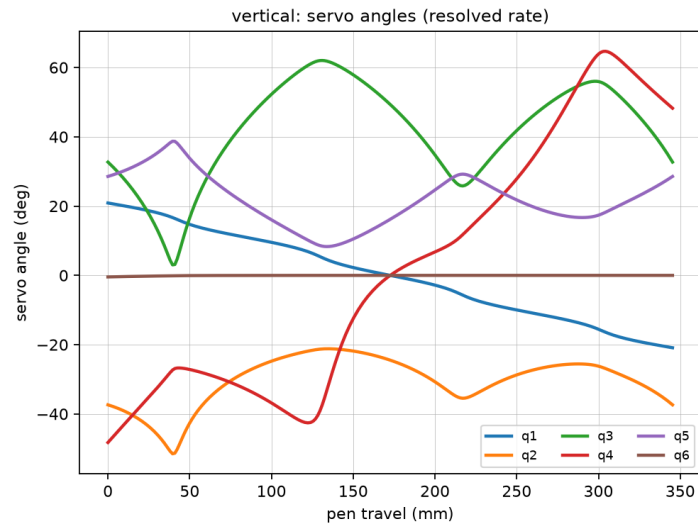
one pass of the sine at half the rating: 0.8 s

encoder floor (every joint off by half a count): tool face $0.84 \dots 0.88 \text{ mm}$ along the path

-> drawing faster than about half the no-load pen speed above leaves the servos no margin (the rating is no-load; under the arm's own weight they are slower). The encoder alone limits a drawn line to about the floor above -- a real line is also limited by gear backlash and the printed parts' play.

10 Servo angles, servo speeds and the encoder floor along both sines

Sine as a timed trajectory on the ST3215 servos: resolved rate (roll free), joint speeds vs the rating, encoder floor



11 Conclusions

- * The Modified-DH table of rev I.1 follows from the CAD with the convention's frame rules and reproduces the CAD and the URDF to 1e-13 mm; the velocity-propagation Jacobian matches the geometric form and finite differences.
- * The arm has a spherical wrist, so its IK is closed-form: up to 8 exact branches in about 0.2 ms, no seed. The numerical solvers (DLS, pseudoinverse, Levenberg-Marquardt, ikpy) are exact only from a nearby seed: good for following a path, unreliable for a target anywhere. Use the closed form to start, a numerical solver (or the closed form with branch continuity) along a path.
- * Home (straight up) is singular three ways (shoulder, elbow, wrist): work from the bent ready pose.
- * The joint limits (set by the parts' clearances) cap the pitch at $J2 + J3 + J5 = 172.5$ deg: the pen can never point straight down. Table drawing needs a tilted pen (least tilt ~ 15 deg near the table) and happens on a ring round the base; a vertical board works with the pen pointing away from the robot, from ~ 190 mm out.
- * Both sines were traced smoothly by using the free roll of a symmetric pen (largest joint step < 3 deg) -- the same redundancy idea as the thesis's raised-elbow branch.
- * Timed on the ST3215 (270 deg/s no-load): the board sine can run at ~ 82 mm/s pen speed with the servos at half their rating; the encoder (0.088 deg) alone sets a ~ 0.93 mm floor on drawing accuracy -- backlash and the printed parts' play come on top.